

Task 5 – IP Addressing Plan

Tech Career Accelerator Week 2
Deliverable

1. Overview

A structured IP addressing scheme is fundamental to network management, troubleshooting, and security segmentation. This plan assigns dedicated subnets to each device category, reducing broadcast traffic, simplifying firewall rule creation, and enabling clear identification of traffic by type. All subnets are drawn from the RFC 1918 private address space and are not routable on the public internet.

2. Subnet Allocation

Device Category	Subnet	Usable Range	Hosts	Notes
Servers	192.168.10.0/24	192.168.10.1 – 192.168.10.254	254	Static IPs; DHCP disabled
User Devices	192.168.20.0/24	192.168.20.1 – 192.168.20.254	254	DHCP pool: .50 – .200
Printers	192.168.30.0/24	192.168.30.1 – 192.168.30.254	254	Static IPs recommended
Wi-Fi / BYOD	192.168.40.0/24	192.168.40.1 – 192.168.40.254	254	Isolated VLAN; NAT to WAN
Management VLAN	192.168.100.0/24	192.168.100.1 – 192.168.100.254	254	Network devices only

Table 1 – Subnet allocations by device category.

3. Reserved Addresses per Subnet

Within each /24 subnet, the following addresses are reserved by convention and must not be allocated to end devices:

Address (last octet)	Purpose
.0	Network address (not assignable)
.1	Default gateway (router interface)
.2	Secondary gateway or HSRP/VRRP virtual IP (where applicable)
.10	DNS server primary
.11	DNS server secondary
.255	Broadcast address (not assignable)

Table 2 – Reserved addresses within each subnet.

4. DHCP Configuration Summary

Subnet	DHCP Enabled	Pool Range	Lease Time
192.168.10.0/24	No	N/A (static only)	N/A
192.168.20.0/24	Yes	192.168.20.50 – 192.168.20.200	8 hours
192.168.30.0/24	No	N/A (static only)	N/A
192.168.40.0/24	Yes	192.168.40.50 – 192.168.40.200	4 hours
192.168.100.0/24	No	N/A (static only)	N/A

Table 3 – DHCP configuration per subnet.

5. DNS and Gateway Configuration

- Primary DNS: 192.168.10.10 (internal resolver — forwards to 1.1.1.1 and 8.8.8.8 for external resolution).
- Secondary DNS: 192.168.10.11 (redundant internal resolver).
- Each subnet's default gateway is the .1 address of that subnet, configured on the core router/firewall.
- Inter-VLAN routing is handled by the Layer 3 switch or firewall; the Wi-Fi/BYOD VLAN (192.168.40.0/24) has no direct route to the Server or Management VLANs.

6. Governance

- All static IP assignments must be recorded in the IP Address Management (IPAM) register before the device is connected.
- No device may be assigned an IP address outside of the subnets defined in this plan without a change request.
- DHCP leases are logged and retained for 90 days to support incident investigation.
- The addressing plan is reviewed annually or when a significant change to the network topology is proposed.